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MSc(IT) Semester II [Winter 2025] $23 \times 2 = 35$

IT632: Software Engineering

Duration: 1 hour

Max. Marks: 40

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23

- ① According to the Layered Approach to Software Engineering, which specific layer is responsible for providing the technical "how-to" details, such as design specifications, coding, and testing? (1 Mark)
- (A) The Process layer, which establishes the framework for management and control.
 - ✓(B) The Methods layer, which encompasses the technical tasks for building software.
 - (C) The Tools layer, which provides the automated support for the development lifecycle.
 - (D) The Quality Focus layer, which serves as the foundational bedrock for the entire process.
2. The Rapid Application Development (RAD) model is characterized by its high-speed execution. What is the typical timeframe and primary emphasis of this model? (1 Mark)
- (A) A 30-day "Sprint" cycle emphasizing self-organizing teams.
 - ✓(B) A 60-90 day cycle emphasizing extensive re-use of components.
 - (C) A continuous cycle emphasizing formal models and automatic code generation.
 - (D) An indefinite cycle emphasizing risk analysis and customer communication.
3. While both the Prototyping Model and the Incremental Model involve iterative development, what is the fundamental technical distinction between their outputs? (1 Mark)
- (A) The Prototyping model delivers a series of working versions, whereas the Incremental model delivers a single "first system" mock-up.
 - (B) The Prototyping model uses a waterfall style for each iteration, whereas the Incremental model avoids waterfall techniques entirely.
 - ✓(C) The Incremental model delivers a working version with every increment, while the Prototyping model typically involves discarding the mock-up to build the product from scratch.
 - (D) The Prototyping model is used only for large-scale systems, whereas the Incremental model is specifically designed for small teams with less than five members.
4. In the Scrum methodology, specific activities are strictly partitioned into phases. Which of the following tasks is exclusively associated with the "Post-game" phase? (1 Mark)
- (A) Creation of the Product Back-log List and prioritizing requirements.
 - (B) Evolution of the system architecture through iterative Sprints.
 - ✓(C) Integration testing, system testing, and documentation preparation.
 - (D) High-level design and the definition of the next sprint's goals.

5. Within the Component-Based Development model, the process is divided into two parallel activities. What is the specific purpose of "Domain Engineering" in this context? (1 Mark)
- (A) Selecting an architectural style to meet specific project objectives.
 - ✓(B) Exploring an application domain to find reusable functional and behavioral components.
 - (C) Modifying existing components to ensure they fit into the new system architecture.
 - (D) Automating the generation of test cases and documentation for the sub-systems.
6. Software process models differ in their suitability for specialized systems. Which of the following statements correctly identifies a limitation or requirement of a specific model? (1 Mark)
- (A) The RAD model is highly recommended for safety-critical systems due to its rigorous waterfall sub-stages.
 - (B) The Concurrent development model is often insisted upon by safety-critical standards, sometimes requiring two copies of the same system.
 - ✓(C) The Linear Sequential Model is the most appropriate choice for projects where requirements change frequently.
 - (D) The Spiral model is unique because it is the only model that does not require customer communication during its passes.
7. A development team is working with a customer who has a general idea of the software's objectives but cannot define the specific functional details or user interface requirements. To avoid building a final product that the customer rejects, which approach should the team apply to refine these requirements jointly? (1 Mark)
- (A) The Linear Sequential Model, to ensure every stage is documented and reviewed.
 - ✓(B) The Prototyping Model, to build a "first system" for the customer to "test drive" and refine requirements.
 - (C) The Scrum Pre-game phase, to ensure all requirements are frozen in a Product Back-log before development.
 - (D) The Component-Based Development model, to use pre-built components that the customer can choose from a catalog.
8. A software firm is tasked with developing a large-scale system where different modules can be developed at the same time. However, within each module, the tasks must follow a specific order. Which model is "very useful" for this scenario, provided there is high clarity during the planning phase? (1 Mark)
- ✓(A) The Concurrent Development Model, where tasks within an entity are serial, but the entities themselves operate simultaneously.
 - (B) The Waterfall Model, by applying its linear stages to the entire large-scale system at once.
 - (C) Extreme Programming (XP), by using Pair Programming to ensure two people work on every line of code.
 - (D) The Model-Based Development approach, to automatically generate the code for all concurrent modules.

9. In an Extreme Programming (XP) environment, the team has finished the "Planning Phase" and is moving into "Iterations to Release." If a customer decides to add a new "Story" (feature) midway through an iteration, how should the team technically handle this according to XP principles? (1 Mark)
- (A) Stop the current iteration immediately and return to the Exploration Phase.
 - (B) Implement the story immediately by bypassing the functional tests.
 - ☒ (C) The customer decides which stories are in each iteration; the change can be implemented in a subsequent iteration or handled during the maintenance phase.
 - ☒ (D) Wait until the Death Phase to write the documentation for the new story.
10. When distinguishing between Functional Requirements (FRs) and Non-Functional Requirements (NFRs), which of the following is true regarding their "End Result"? (1 Mark)
- (A) FRs define product properties, while NFRs define product features.
 - ☒ (B) FRs define product features, while NFRs define product properties.
 - (C) FRs focus on user expectations, while NFRs focus on user requirements.
 - ☒ (D) FRs are generally defined by developers, while NFRs are defined by users.
11. Which of the following is considered a "desirable attribute" of a good systems analyst during the requirements gathering phase? (1 Mark)
- (A) Technical proficiency in high-level coding and database normalization.
 - ☒ (B) Imagination, creativity, and strong interaction skills.
 - ☒ (C) The ability to provide "how-to" implementation solutions immediately.
 - (D) Strict adherence to manual accounting formats without adaptation.
12. An SRS document is said to suffer from "Over-specification" when it does which of the following? (1 Mark)
- (A) Includes information that is irrelevant to the specific problem, also known as "Noise".
 - (B) Omits aspects of the problem that are essential for a proper solution, known as "Silence".
 - ☒ (C) Addresses "how to" aspects of the solution, such as defining specific data structures or sorting orders.
 - (D) Uses literary expressions or unquantifiable terms like "fantastic processing".
13. How does the SRS document function as a "Contract" between the development team and the customer? (1 Mark)
- (A) It serves as a design manual that the development team can update unilaterally as technical constraints arise.
 - (B) It specifies the exact hardware and coding language to be used to ensure the product is legally compliant.
 - ☒ (C) It provides a reference for settling subsequent controversies, ensuring the final product is acceptable if it satisfies the recorded requirements.
 - (D) It is a "Statement of User Needs" that is discarded once the implementation phase begins.

14. Which statement best describes "Verifiability" in the context of a good SRS? (1 Mark)
- (A) It refers to "Wishful Thinking," where the system aims for impossible performance metrics.
 - ☒ (B) It means a requirement is stated such that a finite set of test steps can be written to validate its behavior.
 - (C) It implies that the requirement uses qualitative goals like being "user-friendly" to ensure customer satisfaction.
 - (D) It is the process of ensuring every requirement is linked to a specific developer's name in the revision history.
15. According to the impact levels defined for requirements risk, a "High" risk level is assigned when: (1 Mark)
- (A) It will ~~break~~ the main functionality, rendering the system unusable.
 - ☒ (B) It will impact main functionality and make some functions inaccessible, though the system remains generally usable.
 - (C) It impacts minor features but leaves the main functionality and accessibility untouched.
 - (D) The system can be used without limitations as long as workarounds are provided.
16. A software requirement in the "Security" section states that "Passwords must be exactly 8 characters long," while a requirement in the "User Interface" section of the same document states that "The password field shall accept between 8 and 16 characters." Which quality goal is primarily violated here? (1 Mark)
- (A) Ambiguity: The requirement is written using vague ranges that cannot be verified by a tester.
 - ☒ (B) Inconsistency: Two parts of the specification provide contradictory constraints for the same system attribute.
 - (C) Over-specification: The analyst is incorrectly telling the designer "how" to store the password instead of "what" to store.
 - (D) Silence: The document fails to define the character set (alphanumeric vs. special characters) for the password.
17. Consider the requirement: "The system shall calculate the daily tax liability for every transaction and update the ledger accordingly." Based on the standard SRS functional template, how should this be classified? (1 Mark)
- (A) ~~Non-functional Requirement (Performance)~~: Because it involves a calculation that must be done daily to satisfy user expectations.
 - ☒ (B) Processing / System Behavior: It defines a mandatory system action that transforms input data into an observable result.
 - (C) Goal of Implementation: It is a desirable property that would not be checked for strict compliance during the testing phase.
 - (D) External Interface Requirement: It describes how the software interacts with the hardware's mathematical processor.

- According to the "3 Cs" framework, what is the primary role of the "Confirmation" component in a User Story? (1 Mark)
- (A) It serves as a legal sign-off between the Product Owner and the Developers to prevent scope creep.
 - ✓ (B) It represents acceptance tests that verify the story has been coded correctly according to the conditions of satisfaction.
 - (C) It is the final documentation updated in the data dictionary after a feature is deployed.
 - ✓ (D) It is the process where the Scrum Master confirms that the team has the capacity to finish the story.
19. What is the recommended size constraint for an individual requirement to be considered "appropriately sized" for an increment? (1 Mark)
- (A) It should be no shorter than one ideal day and no longer than the Sprint duration.
 - (B) It must be small enough to be completed in a Daily Scrum but large enough to provide customer value.
 - ✓ (C) It should be no shorter than half a day and no longer than the length of an increment.
 - (D) It must be exactly one-thirteenth of the total velocity based on the Fibonacci sequence.
20. Story points are calculated by aggregating three specific factors. Which of the following best describes these factors? (1 Mark)
- (A) Individual developer speed, historical project data, and the total number of sub-tasks.
 - ✓ (B) The amount of work, the complexity of the logic, and the degree of uncertainty or risk involved.
 - (C) The ratio of ideal days to actual hours, adjusted by the team's current velocity.
 - (D) Client priority, the Fibonacci weight, and the documented technical debt.
21. Why is the common practice of writing stories from the perspective of "The User" considered a mistake in Agile requirements gathering? (1 Mark)
- (A) Because "The User" is too technical a term and shifts the focus from the business goal to system attributes.
 - ✓ (B) It assumes a homogenous group, leading the team to overlook the varying backgrounds, goals, and roles of different stakeholders.
 - (C) It violates the Agile Manifesto principle of prioritizing "Individuals and interactions" over "Processes and tools".
 - (D) It makes it impossible for the Product Owner to prioritize the backlog based on risk profile.
22. According to the official Scrum Guide, what is the consequence of adopting only specific parts of the Scrum framework? (1 Mark)
- ✓ (A) It is considered "Scrum-Lite," which is recommended for teams new to the methodology.
 - ✗ (B) The team achieves "Adaptive Behavior" but loses the benefits of time-boxing.
 - (C) The result is not Scrum, as the framework is immutable and each element serves a specific purpose.
 - (D) It allows the team to focus on Technical Implementation rather than Empirical Processes.

23. A team is estimating a story and cannot decide if it should be an 8 or a 13 on the Fibonacci scale. Based on Agile theory, what is the most appropriate action? (1 Mark)
- (A) Take the average (10.5) to ensure the Burn Down Chart remains accurate.
- (B) Default to 8 points to avoid over-commitment during the Sprint Planning.
- ☒ (C) Treat the story as too large or uncertain and consider breaking it down into smaller, more manageable stories.
- (D) Assign it T-shirt size "Large" to avoid the "false precision" of numerical estimates.
24. You are developing a reservation system. The story "As a user, I can cancel a reservation" is found to have complex rules (e.g., different refund rates for premium members). How should these details be handled? (1 Mark)
- (A) They must be documented in a separate Data Dictionary or Annex to keep the story card "short".
- ☒ (B) They should be added as Conditions of Satisfaction (tests) or split into sub-stories.
- (C) The story should be converted back into a Waterfall-style requirement to ensure no details are overlooked.
- (D) The Scrum Master should decide the rules during the Daily Scrum to maintain team synchronization.
25. A new software project has a total development timeline of five months. Based on Agile recommendations for project duration and routine, what is the most suitable increment length? (1 Mark)
- (A) 30 days, as this is the standard duration recommended by the SCRUM framework to handle complexity.
- ☒ (B) Two-week increments, because the total project timeline is less than 6 months.
- (C) Variable lengths (1-4 weeks) depending on whether the increment contains an Epic or a Sub-story.
- (D) A one-week increment for the first month to allow for sufficient preparation time and analysis.
26. How are software programs different from software products? (1.5 Marks)

Ans:	Software Programs	Software Products
	- Small in size - Used by the sole developer - Does not have proper user interface	- Large in size - Used by a team - Have a proper user interface

1.5

7. Differentiate giving suitable examples - stakeholders vs users for software projects.

(1.5 Marks)

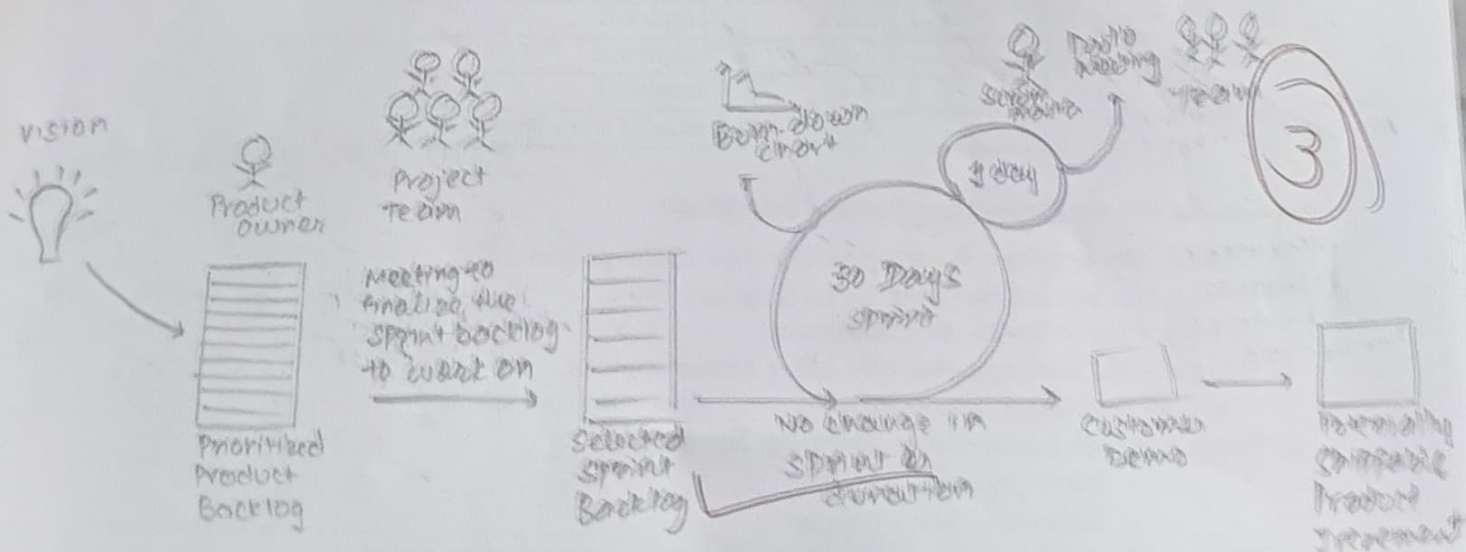
Ans: Stakeholders may play some role in the software project but are not necessarily the users.

- While software users are the people who are actually going to use the software.
- All software users are stakeholders but not all stakeholders are software users.
- Eg: In a restaurant management system software
 - cashier is the user of software as well as have some stake
 - Hotel owner may not directly use the software but is the stakeholder of it.

28. Visualize a model that logically connects the following terms (and any related missing terms). Your diagram should have at minimum all the following terms and more (if any).

Sprint, Daily Meeting, Selected Sprint Backlog, Customer Demo, Potentially Shippable Product Increment, Burn-down Chart, Vision, Prioritized Product Backlog, Product Owner, Project Team etc.

(3 Marks)



Correct!!!

20. A software development team is building a **Software System** using **Scrum**. The product owner has prepared a set of user stories with estimated story points. The project has already completed three sprints, and partial progress data is available. Assume story points are burned only when a story is fully completed. The project context details are given in the following. (5 Marks)

- **Sprint duration:** 2 weeks
- **Total planned sprints:** 6
- **Team size:** 5 developers
- **Sprint capacity assumption:** Fixed team, no planned leaves
- **Estimation technique:** Story points (Fibonacci scale)

1 3 5 8 13 21 34
13 5 8
1 1 2 3 5 8 13 20 33 53

Product Backlog (Initial)

Story ID	User Story (Summary)	Story Points
US-01	Visitor pre-registration via mobile app	8
US-02	QR code generation for visitors	5
US-03	Security guard check-in/check-out screen	13
US-04	Resident approval notification	5
US-05	Visitor analytics dashboard	8
US-06	Emergency lockdown broadcast feature	13
US-07	Multi-floor apartment configuration	8
	Total estimated effort	60

Sprint Planning and Execution Data

Sprint	Story ID	Status at Sprint End
1	US-01	Completed $8 > 13$
1	US-02	Completed $5 > 13$
1	US-04	Not completed
2	US-04	Completed $5 > 13$
2	US-03	Completed 13
3	US-05	Completed $8 > 8$
3	US-06	Partially completed

Answer the following showing suitable calculations:

1. Identify the **product backlog** at the start of the project. 60
2. Identify the **sprint backlog** for each sprint. 20
3. Compute the **velocity** of each sprint. $13 + 18 + 8/3 = 39/3 = 13$
4. Compute the **average velocity** after **three sprints**.
5. Compute committed and completed story points in every sprint. 5
6. Draw a **sprint burn-down chart** for **Sprint 3**.
 - X-axis: Days (10 working days)
 - Y-axis: Remaining story points
7. Draw a **release burn-down chart** for the entire project after Sprint 3.

8. Calculate remaining story points. 21
9. Estimate the number of additional sprints required to complete the project.
10. Determine whether the project is: Ahead of schedule, On track, or At risk.

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Ans:

1. Product Backlog : 60 US-01, US-02, US-04, US-03, US-05, US-06, US-07

2. Sprint Backlog :

1] US-01, US-02, US-04

4] US-06, US-07

2] US-03

3] US-05, US-06

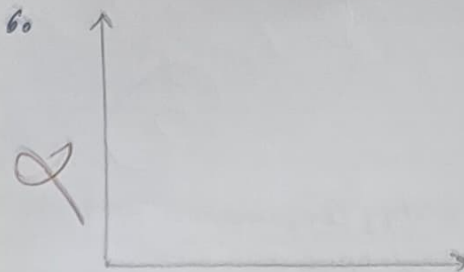
3. Velocity of each sprint = $60/6 = 10$

4. Avg velocity after 3 sprints = $13 + 18 + 8/3 = 39/3 = 13$

5. Sprint 1 : Committed +18
Completed -13

Sprint 2 : Committed -13
Completed -18

Sprint 3 : Committed -21
Completed -8



8. Remaining story points : 21

9. 2

10. Ahead of schedule

30. Given specification: When a customer removes an item from the cart, the total price must immediately reflect the updated contents of the cart. Write requirements specifications containing details about ReqID, Name, Actor, Input, Output and Processing. (2 Marks)

Ans:

ReqID : FR-01

Name : Cart must be updated on removing an item

Actor : User

Input : Selecting an item to remove from cart

Output : Selected item is removed from the cart

Processing : System should remove the item selected by user from the cart and update the cart value & items

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31. Given specification: The system should notify the homeowner whenever daily energy consumption exceeds a user-defined threshold. Write requirements specifications containing details about ReqID, Title, Description, Priority, and Risk. (2 Marks)

Ans:

ReqID : FR-01

Title : Warning user on exceeding the daily energy consumption threshold

Description : The user must receive an warning/notification from system on surpassing the threshold set by him/her for daily energy consumption.

Priority : High

Risk : 80%.

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